Lecture 18

The Annihilator Method and Higher-Order Variation of Parameters

We want to find a method to solve non-homogeneous constant coefficient equations of higher order such as $D(y) = y'$, $D^n(y) = y^{(n)}$

$$a_n y^{(n)} + \cdots a_1 y' + a_0 = f(t)$$

which can be expressed as

$$\underbrace{[a_n D^n + \cdots a_1 D + a_0]}_{\text{char eqn}}(y) = f(t)$$

using the differential operator D.

When $f(t)$ had a suitable form for 2nd-order equations we developed the method of undetermined coefficients. The forms of $f(t)$ that we could apply that method for were:

- polynomials t^m
- exponentials e^{rt} $e^{\alpha t}$
- sine or cosine $\sin(\beta t)$ $\cos(\beta t)$
- linear combinations of products of these functions $te^{\alpha t}\cos(\beta t)$

$f(t)$

Now that we have more experience with solving constant coefficient equations, one may note that all of these suitable types of functions are themselves the solution types for homogeneous constant coefficient equations.

That is, they all solve a homogeneous constant coefficient equation. In fact, we can write down each of the equations they solve using the differential operator D.

- polynomials and exponentials - i.e. $f(t) = t^k e^{rt}$ for $k = 0, 1, \ldots, m-1$.

  m - multiplicity of root r $\quad y_h = c_1 e^{rt} + c_2 t e^{rt} + \cdots + c_m t^{m-1} e^{rt}$

  This solves the equation $[(D-r)^m](y) = 0$ and we can write $[(D-r)^m](f) = 0$.

  $[(D-1)^2](y) = 0 \quad y_h = c_1 e^t + c_2 t e^t$

  $y'' - 2y' + y = 0$

- sine and cosine - i.e. $f(t) = \sin(\beta t)$ or $f(t) = \cos(\beta t)$.

  $y'' + \beta^2 y \qquad r = i\beta \rightarrow D^2 + \beta^2$

  This solves the equation $[D^2 + \beta^2](y) = 0$ and we can write $[D^2 + \beta^2](f) = 0$.

- A combination of these two - i.e. $f(t) = t^k e^{\alpha t} \cos(\beta t)$ or $f(t) = t^k e^{\alpha t} \sin(\beta t)$.

  $r = \alpha + i\beta$

  This solves the equation $[((D-\alpha)^2 + \beta^2)^m](y) = 0$ and we can write $[((D-\alpha)^2 + \beta^2)^m](f) = \boxed{0}$.

So we note that by applying each of these expressions involving the operator D, the function $f(t)$ evaluates to zero. We say that the function $f(t)$ is **annihilated** by the appropriate operator expression.

Definition: A linear differential operator A is said to **annihilate** a function $f(t)$ if

expression in D

$$[A](f(t)) = 0$$

for all t. That is, A annihilates f if f is a solution to the above homogeneous equation.

$y_h = c_1 y_1 + c_2 y_2 + \cdots$

Example: Write an expression involving the differential operator D that annihilates

$$\underbrace{8t^2e^{-2t}}_{1} + \underbrace{4e^t\sin(2t)}_{2}$$

$m = \text{power} + 1$
$2 + 1 = 3$

① : $8t^2\underbrace{e^{-2t}}_{r=-2} \rightarrow (D+2)^3 \Rightarrow [(D+2)^3](8t^2e^{-2t}) = 0$ ✓

no t's

② : $4\underbrace{e^t\sin(2t)}_{r=1+2i}$ $((D-1)^2+2^2) \Rightarrow [(D-1)^2+4](4e^t\sin(2t)) = 0$ ✓

Compose the operator expressions to annihilate both functions

$$[(D+2)^3]([(D-1)^2+4](8t^2e^{-2t}+4e^t\sin(2t))) = 0$$

The operator expression $(D+2)^3((D-1)^2+4)$ annihilates $8t^2e^{-2t}+4e^t\sin(2t)$

If we suppose that there is a linear differential operator A (expression in D) that annihilates a function $f(t)$, then we can express a non-homogeneous constant coefficient equation with forcing function $f(t)$

$$[A]([a_nD^n + \cdots a_1D + a_0](y) = f(t)) \tag{18.1}$$

as a homogeneous constant coefficient equation by applying the operator A to both sides as

$$[A]([a_nD^n + \cdots a_1D + a_0](y)) = [A](f(t)) = 0 \tag{18.2}$$

and so any solution to (18.1) must also be a solution to (18.2).

New homogeneous eqn: $[A](a_nD^n + \cdots a_1D + a_0](y)) = 0$

We can explore this idea in the following example:

Example: Consider the equation

$$y'' - y = te^t + \sin(t)$$

$D^2y - y$

If we rewrite the equation in operator form we have $(D-1)^2$

$m=2$ $r=1$

$$[D^2 - 1](y) = te^t + \sin(t).$$

$r=i$ D^2+1

Both te^t and $\sin(t)$ are annihilated by the operators $(D-1)^2$ and D^2+1, respectively. Therefore we can write the equation as a homogeneous equation by applying these operators to both sides successively and obtain

$$\begin{aligned}
[(D-1)^2]([D^2+1]([D^2-1](y))) &= [(D-1)^2]([D^2+1](te^t + \sin(t))) \\
&= [D^2+1]([(D-1)^2](te^t)) + [(D-1)^2]([D^2+1](\sin(t))) \\
&= [D^2+1](0) + [(D-1)^2](0) \\
&= 0.
\end{aligned}$$

If we now write the resulting homogeneous equation we have

$$[(D-1)^2(D^2+1)(D^2-1)](y) = 0 \quad \Rightarrow \quad \underbrace{[(D+1)(D-1)^3(D^2+1)](y) = 0}_{r=-1 \quad r=1 \quad r=i}$$

$(D-1)(D+1)$

This is a homogeneous constant coefficient equation and we can immediately write down the solution as

$$y = \underbrace{C_1e^{-t} + C_2e^t}_{y_h} + \overbrace{C_3te^t + C_4t^2e^t + C_5\sin(t) + C_6\cos(t)}^{y_p}$$

The function y must also satisfy the original equation $y'' - y = te^t + \sin(t)$ and so we only need to substitute back into the original equation to find the appropriate values for the constants. Of note is that $C_1e^{-t} + C_2e^t$ is exactly the solution to the associated homogeneous equation $y'' - y = 0$ and so we

$y'' - y = te^t + \sin(t)$

only need to determine the constants C_3, C_4, C_5, and C_6 to find the particular solution.

$$y_p = C_3 te^t + C_4 t^2 e^t + C_5 \sin(t) + C_6 \cos(t)$$

$$y_p' = C_3(e^t + te^t) + C_4(2te^t + t^2e^t) + C_5 \cos(t) - C_6 \sin(t)$$

$$y_p'' = C_3(e^t + e^t + te^t) + C_4(2e^t + 2te^t + 2te^t + t^2e^t) - C_5 \sin(t) - C_6 \cos(t)$$

$$y'' - y = (C_3(2e^t + \cancel{te^t}) + C_4(2e^t + 4te^t + \cancel{t^2e^t}) - C_5 \sin(t) - C_6 \cos(t))$$
$$- (\cancel{C_3 te^t} + \cancel{C_4 t^2 e^t} + C_5 \sin(t) + C_6 \cos(t)) = te^t + \sin(t)$$

$$\Rightarrow e^t(2C_3 + 2C_4) + te^t(4C_4) + \sin(t)(-2C_5) + \cos(t)(-2C_6) = te^t + \sin(t)$$

$$\begin{cases} 2C_3 + 2C_4 = 0 \Rightarrow 2C_3 + \frac{1}{2} = 0 \Rightarrow C_3 = -\frac{1}{4} \\ 4C_4 = 1 \Rightarrow C_4 = \frac{1}{4} \\ -2C_5 = 1 \Rightarrow C_5 = -\frac{1}{2} \\ -2C_6 = 0 \Rightarrow C_6 = 0 \end{cases}$$

so the general solution is

$$\boxed{y_g = C_1 e^{-t} + C_2 e^t - \frac{1}{4} te^t + \frac{1}{4} t^2 e^t - \frac{1}{2} \sin(t)}$$

In fact, if we had used undetermined coefficient we would have come up with the same function as the general solution. Notice how the annihilators included the need to multiply by successive powers of t in repeated roots.

This method of using annihilators to rewrite our non-homogeneous equation as a homogeneous one is called the **Annihilator Method**.

Example: Use the annihilator method to solve

$$y''' + y'' - 5y' + 3y = e^{-t} + \sin(t)$$

$$\Rightarrow [D^3 + D^2 - 5D + 3](y) = \underbrace{e^{-t}}_{1} + \underbrace{\sin(t)}_{2}$$

① : e^{-t} $r = -1$, $D+1$

② : $\sin(t)$ $r = i$, D^2+1

$$\Rightarrow [(D^2+1)(D+1)(\underbrace{D^3+1)^2-5D+3}_{\text{factor}})](y) = 0$$

1 3

rational roots theorem

$$r = \pm \frac{\text{Factor of constant}}{\text{factor of leading}}$$

$$= \pm 1, \pm 3$$

$D=1$: $1^3 + 1^2 - 5(1) + 3 = 5 - 5 = 0$ ✓

$$\begin{array}{r|l} & D^2 + 2D - 3 \\ \hline D-1 & D^3 + D^2 - 5D + 3 \\ & -(D^3 - D^2) \\ & \quad +2D^2 - 5D \\ & \quad -(2D^2 - 2D) \\ & \qquad -3D + 3 \\ & \qquad -(-3D + 3) \\ & \qquad\quad 0 \end{array}$$

$$[(D^2+1)(D+1)(D-1)(D^2+2D-3)](y) = 0$$

$$\Rightarrow [(D^2+1)(D+1)(D-1)(D+3)(D-1)](y) = 0$$

$$\Rightarrow [(D^2+1)(D+1)(D-1)^2(D+3)](y) = 0$$

$r = i$ $r = -1$ $r = 1$, $m = 2$ $r = -3$

So $y_g = \underbrace{(c_1 e^{t} + c_2 t e^{t} + c_3 e^{-3t})}_{y_h} + \underbrace{c_4 e^{-t} + c_5 \cos(t) + c_6 \sin(t)}_{y_p}$

Need to substitute to find c_4, c_5, and c_6.

$$y''' + y'' - 5y' + 3y = e^{-t} + \sin(t)$$

$$y_p = c_4 e^{-t} + c_5 \cos(t) + c_6 \sin(t)$$

$$y_p' = -c_4 e^{-t} - c_5 \sin(t) + c_6 \cos(t)$$

$$y_p'' = c_4 e^{-t} - c_5 \cos(t) - c_6 \sin(t)$$

$$y_p''' = -c_4 e^{-t} + c_5 \sin(t) - c_6 \cos(t)$$

So

$$(-\cancel{c_4 e^{-t}} + c_5 \sin(t) - c_6 \cos(t)) + (\cancel{c_4 e^{-t}} - c_5 \cos(t) - c_6 \sin(t))$$
$$- 5(-c_4 e^{-t} - c_5 \sin(t) + c_6 \cos(t)) + 3(c_4 e^{-t} + c_5 \cos(t) + c_6 \sin(t))$$
$$= e^{-t} + \sin(t).$$

$$\Rightarrow e^{-t}(8c_4) + \cos(t)(2c_5 - 6c_6) + \sin(t)(6c_5 + 2c_6) = e^{-t} + \sin(t)$$

So
$$\begin{cases} 8c_4 = 1 & \Rightarrow c_4 = \frac{1}{8} \\ 2c_5 - 6c_6 = 0 & \Rightarrow c_5 = 3c_6 \quad \Rightarrow c_5 = \frac{3}{20} \\ 6c_5 + 2c_6 = 1 & \Rightarrow 18c_6 + 2c_6 = 1 \\ & \Rightarrow 20c_6 = 1 \Rightarrow c_6 = \frac{1}{20} \end{cases}$$

Thus
$$\boxed{y_g = (c_1 e^{t} + c_2 t e^{t} + c_3 e^{-3t}) + \frac{1}{8} e^{-t} + \frac{3}{20} \cos(t) + \frac{1}{20} \sin(t)}$$

Now, we turn our attention to solving general linear nth-order equations with forcing functions that do not have to be solutions to homogeneous constant coefficient equations. In the case for 2nd-order equations, we developed the method of variation of parameters to take a homogeneous solution set and construct a particular solution. We will derive a higher order version of this to help solve higher order non-homogeneous equations.

Suppose we wish to find a solution to the linear nth-order equation

functions of t

$$a_n(t)y^{(n)} + \cdots a_1(t)y' + a_0y = f(t)$$

and we know that the $y_h = C_1y_1 + \cdots C_ny_n$ is the solution to the homogeneous equation

$$a_n(t)y^{(n)} + \cdots a_1(t)y' + a_0y = 0.$$

Just as in the 2nd-order case we will seek to find a particular solution of the form

$y_p = v_1y_1 + v_2y_2$

$$y_p(t) = v_1(t)y_1 + \cdots + v_n(t)y_n$$

and then find the appropriate function $v_1, \ldots, v_n$ that force y_p to satisfy the equation. We can find them by substitution and construct a system of equations.

$v_1y_1' + v_2y_2' + (v_1'y_1 + v_2'y_2)$
$= 0$

If we take the first derivative y_p' we obtain

$= 0$

$$y_p' = (v_1y_1' + \cdots v_ny_n') + (v_1'y_1 + \cdots + v_n'y_n)$$

after rearranging each product rule. We impose the restriction $v_1'y_1 + \cdots + v_n'y_n = 0$ so that we do not obtain any higher order derivatives of $v_1, \ldots, v_n$ in our construction. By applying this idea successively to $y_p'', y_p''', \ldots, y_p^{(n-1)}$ we will continue to impose restriction on the equations and obtain the system of

2nd order

$$\begin{cases} v_1'y_1 + v_2'y_2 = 0 \\ v_1'y_1' + v_2'y_2' = \frac{f}{a} \end{cases}$$

n linear equations

$$\begin{cases} v_1'y_1 + \cdots + v_n'y_n = 0 \\ v_1'y_1' + \cdots + v_n'y_n' = 0 \\ \quad\vdots \\ v_1'y_1^{(n-1)} + \cdots + v_n'y_n^{(n-1)} = \frac{f(t)}{a_n(t)} \end{cases}$$

which can be written in matrix form as

$$\underbrace{\begin{bmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{bmatrix}}_{\text{Wronskian}} \cdot \begin{bmatrix} v_1' \\ v_2' \\ \vdots \\ v_n' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \frac{f(t)}{a_n(t)} \end{bmatrix}$$

To solve this we employed Cramer's Rule in the 2nd-order case. We can employ a higher order version of Cramer's Rule (which we won't discuss in detail) and obtain the following solutions:

$$v_i'(t) = \frac{W_i}{W}, \quad i = 1, 2, \ldots, n$$

det $\begin{bmatrix} 0 \\ \vdots \\ f/a \end{bmatrix}$

Wronskian

where $W = W[y_1, y_2, \ldots y_n](t)$ is the Wronskian of $y_1, y_2, \ldots, y_n$ and W_i is the determinant of the Wronskian matrix with the i column replaced by $[0, 0, \ldots, f/a]$ (just as in the 2nd-order case).

Therefore, each v_i can be obtained through integration as

$$v_i(t) = \int \frac{W_i}{W} dt$$

and we can write the particular solution as

analogous to 2nd order

$$y_p(t) = v_1 y_1 + \cdots + v_n y_n.$$

We summarize this in the following:

Theorem (Higher Order Variation of Parameters): Suppose that $y_h = C_1 y_1 + \cdots + C_n y_n$ is a solution to the homogeneous linear nth-order equation

$$a_n(t)y^{(n)} + \cdots + a_1(t)y' + a_0 y = 0.$$

Then a particular solution to the equation

$$a_n(t)y^{(n)} + \cdots + a_1(t)y' + a_0 y = f(t)$$

is given by $y_p = \underline{v_1 y_1 + \cdots + v_n y_n}$ where

$$v_i = \int \frac{W_i}{W} dt$$

with W the Wronskian of $y_1, \ldots, y_n$ and W_i the determinant of the Wronskian matrix with the ith column replaced by $[0, 0, \ldots, f/a]$.

Example: Find a general solution to the equation $f(t) = \tan(t)$

$a = 1$

$$y''' + y' = \tan(t), \quad 0 < t < \frac{\pi}{2}$$

Here $[D^3 + D](y) = \tan(t)$

char eqn: $r^3 + r = 0 \Rightarrow r(r^2+1) = 0$

$r = 0,\ r = i$

$$y_h = c_1 + c_2\cos(t) + c_3\sin(t)$$

$y_1 = 1,\ y_2 = \cos(t),\ y_3 = \sin(t),\ \frac{f}{a} = \tan(t)$

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 1 & \cos(t) & \sin(t) \\ 0 & -\sin(t) & \cos(t) \\ 0 & -\cos(t) & -\sin(t) \end{vmatrix}$$

$$= 1\cdot(-1)^{1+1}\,\text{minor}(W)_{11} + \cos(t)(-1)^{1+2}\,\text{minor}(W)_{12} + \sin(t)(-1)^{1+3}\,\text{minor}(W)_{13}$$

$$= 1(1)\begin{vmatrix} -\sin(t) & \cos(t) \\ -\cos(t) & -\sin(t) \end{vmatrix} + \cos(t)(-1)\begin{vmatrix} 0 & \cos(t) \\ 0 & -\sin(t) \end{vmatrix} + \sin(t)\begin{vmatrix} 0 & -\sin(t) \\ 0 & -\cos(t) \end{vmatrix}$$

$$= 1\underbrace{(\sin^2(t)+\cos^2(t))}_{\to 1} + \cos(t)(-1)(0-0) + \sin(t)(0-0)$$

$$= 1$$

$$W_1 = \begin{vmatrix} 0 & \cos(t) & \sin(t) \\ 0 & -\sin(t) & \cos(t) \\ \tan(t) & -\cos(t) & -\sin(t) \end{vmatrix} = \cdots = \tan(t)$$

$$W_2 = \begin{vmatrix} 1 & 0 & \sin(t) \\ 0 & 0 & \cos(t) \\ 0 & \tan(t) & -\sin(t) \end{vmatrix} = \cdots = -\cos(t)\tan(t) = -\sin(t)$$

$$W_3 = \begin{vmatrix} 1 & \cos(t) & 0 \\ 0 & -\sin(t) & 0 \\ 0 & -\cos(t) & \tan(t) \end{vmatrix} = \cdots = -\sin(t)\tan(t)$$

$$v_1 = \int \frac{W_1}{W}\,dt = \int \frac{\tan(t)}{1}\,dt = \int \tan(t)\,dt = \ln|\sec(t)|$$

$$v_2 = \int \frac{W_2}{W}\,dt = \int \frac{-\sin(t)}{1}\,dt = \int -\sin(t)\,dt = \cos(t)$$

$$v_3 = \int \frac{W_3}{W}\,dt = \int \frac{-\sin(t)\tan(t)}{1}\,dt = \int \frac{-\sin^2(t)}{\cos(t)}\,dt$$

$$= \int -\frac{(1-\cos^2(t))}{\cos(t)}\,dt$$

$$= \int -\sec(t) + \cos(t)\,dt = \underbrace{-\ln|\sec(t)+\tan(t)| + \sin(t)}_{v_1y_1 + v_2y_2 + v_3y_3}$$

$$y_g = y_h + y_p = c_1 + c_2\cos(t) + c_3\sin(t) + (\ln|\sec(t)| + \cos^2(t) + (-\ln|\sec(t)+\tan(t)| + \sin(t))\sin(t))$$

$$\boxed{y_g = c_1 + c_2\cos(t) + c_3\sin(t) + \ln|\sec(t)| - \ln|\sec(t)+\tan(t)|\sin(t)}$$

$at^3y''' + bty' + cy = 0$

Example: Find a general solution to the Cauchy-Euler equation

$$t^3y''' + t^2y'' - 2ty' + 2y = t^3\sin(t), \quad t > 0$$

$a=1$ $b=1$ $c=-2$ $d=2$

Note: The characteristic equation for a homogeneous 3rd-order C-E equation

$$at^3y''' + bt^2y'' + cty' + d = 0$$

is

$$ar^3 + [b - 3a]r^2 + [2a - b + c]r + d = 0$$

char eqn: $r^3 + (1-3)r^2 + (2-1+(-2))r + 2 = 0$

$\Rightarrow r^3 - 2r^2 - r + 2 = 0$

$\Rightarrow r^2(r-2) - 1(r-2) = 0 \Rightarrow (r^2-1)(r-2) = 0$

$(r-1)(r+1)(r-2) = 0 \Rightarrow r = 1, -1, 2$

$y_h = c_1 \underset{y_1}{t} + c_2 \underset{y_2}{t^{-1}} + c_3 \underset{y_3}{t^2}$, $\quad f(t) = t^3\sin(t)$, $a(t) = t^3$

$\frac{f}{a} = \sin(t)$

$$W = \begin{vmatrix} t & t^{-1} & t^2 \\ 1 & -t^{-2} & 2t \\ 0 & 2t^{-3} & 2 \end{vmatrix} = \cdots = -\frac{6}{t}$$

$$W_1 = \begin{vmatrix} 0 & t^{-1} & t^2 \\ 0 & -t^{-2} & 2t \\ \sin(t) & 2t^{-3} & 2 \end{vmatrix} = \cdots = 3\sin(t)$$

$$W_2 = \begin{vmatrix} t & 0 & t^2 \\ 1 & 0 & 2t \\ 0 & \sin(t) & 2 \end{vmatrix} = \cdots = -t^2\sin(t)$$

ZBD

$$W_3 = \begin{vmatrix} t & t^{-1} & 0 \\ 1 & -t^{-2} & 0 \\ 0 & 2t^{-3} & \sin(t) \end{vmatrix} = \cdots = -2t^{-1}\sin(t).$$

$t \Rightarrow \sin$
$1 \Rightarrow -\cos$
$0 \Rightarrow -\sin$

$$v_1 = \int \frac{W_1}{W}\,dt = \int \frac{3\sin(t)}{\frac{-6}{t}}\,dt = \int -\frac{1}{2}t\sin(t)\,dt =$$

$$\overset{IBP}{=} -\frac{1}{2}\left(-t\cos(t) + \sin(t)\right)$$

$$v_2 = \int \frac{W_2}{W}\,dt = \int \frac{-t^2\sin(t)}{\frac{-6}{t}}\,dt = \int \frac{1}{6}t^3\sin(t)\,dt$$

$$\overset{IBP}{=} \frac{1}{6}\left(-t^3\cos(t) + 3t^2\sin(t) + 6t\cos(t) - 6\sin(t)\right)$$

t^3	$\sin$
$3t^2$	$-\cos$
$6t$	$-\sin$
6	$\cos$
0	$\sin$

$$v_3 = \int \frac{W_3}{W}\,dt = \int -\frac{2t^{-1}\sin(t)}{\frac{-6}{t}}\,dt = \int \frac{1}{3}\sin(t)\,dt = -\frac{1}{3}\cos(t)$$

So

$$\boxed{\begin{aligned} y_g = {} & (c_1 t + c_2 t^{-1} + c_3 t^2) \\ & + \left(\tfrac{1}{2}t\cos(t) - \tfrac{1}{2}\sin(t)\right)t \\ & + \left(-\tfrac{1}{6}t^3\cos(t) + \tfrac{1}{2}t^2\sin(t) + t\cos(t) - \sin(t)\right)t^{-1} \\ & + \left(-\tfrac{1}{3}\cos(t)\right)t^2 \end{aligned}}$$